

Digital Image Processing

Lecture # 9 **Image Segmentation**

Image Segmentation

- Group similar components (such as, pixels in an image, image frames in a video)
- Applications: Finding tumors, veins, etc. in medical images, finding targets in satellite/aerial images, finding people in surveillance images, summarizing video, etc.

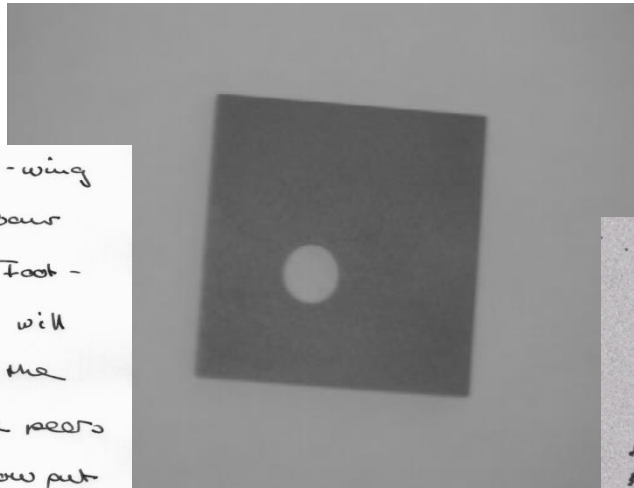
Image Segmentation

- Segmentation algorithms are based on one of two basic properties of gray-scale values:
 - Discontinuity
 - Partition an image based on abrupt changes in gray-scale levels.
 - Detection of isolated points, lines, and edges in an image.
 - Similarity
 - Thresholding, region growing, and region splitting/merging.

Thresholding

- Segmentation into two classes/groups
 - Foreground (Objects)
 - Background

Though they may gather some left-wing support, a large majority of Labour MPs are likely to turn down the Foot-Griffiths resolution. Mr. Foot's line will be that as Labour MPs opposed the Government Bill which brought life peers into existence, they should not now put forward nominees. He believes that the House of Lords should be abolished and that Labour should not take any steps which would appear to "prop up" an out-



Thresholding

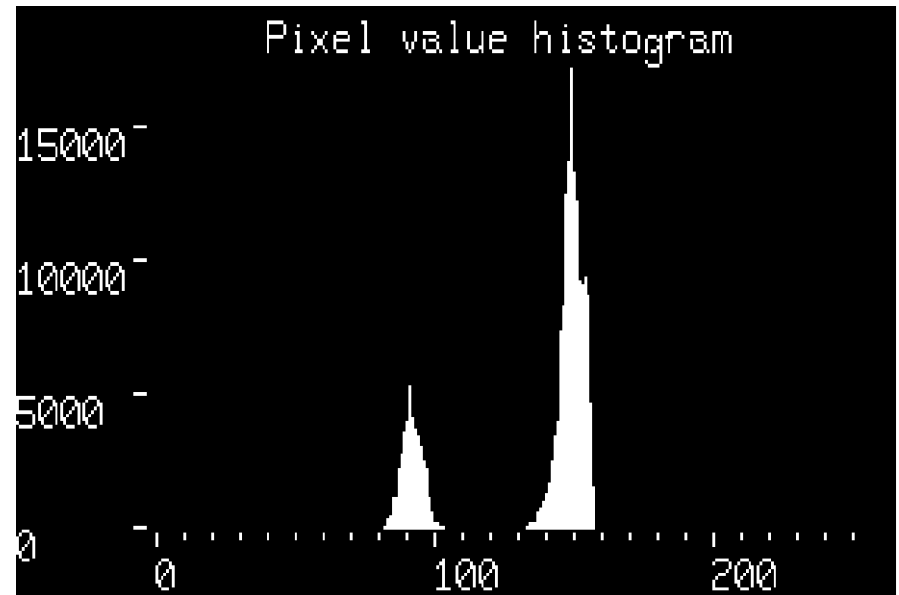
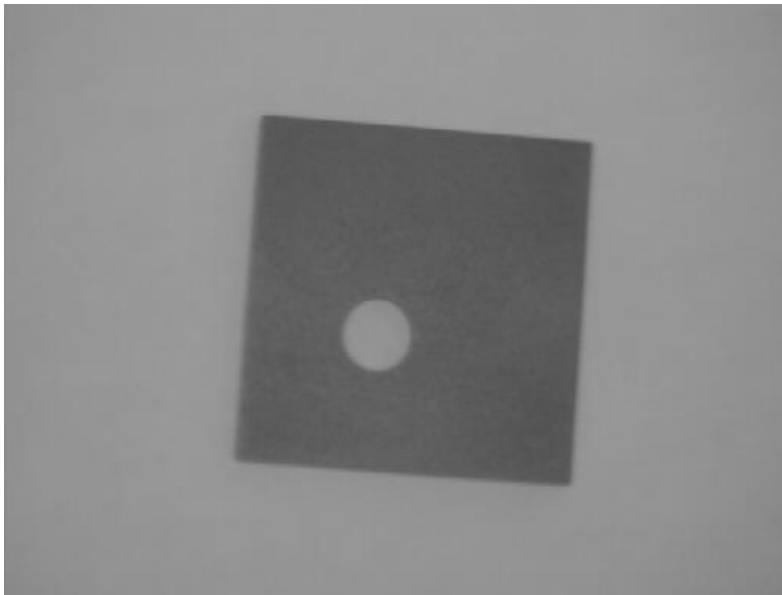
$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases}$$

Objects & Background

- Global Thresholding
- Local/Adaptive Thresholding

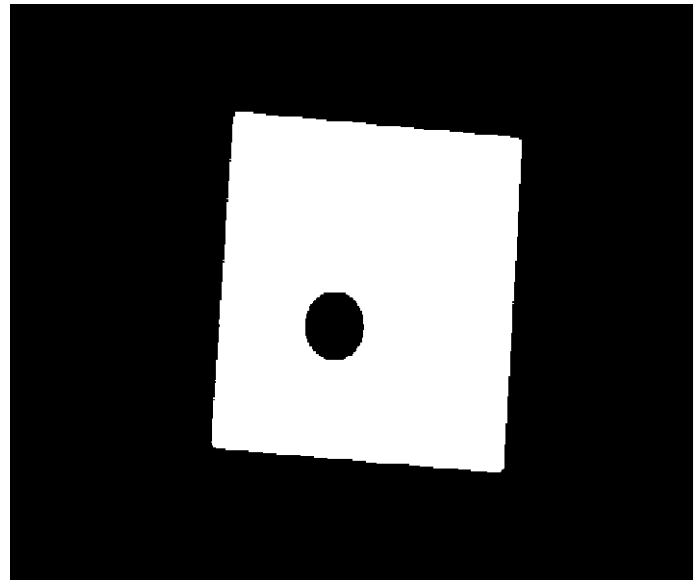
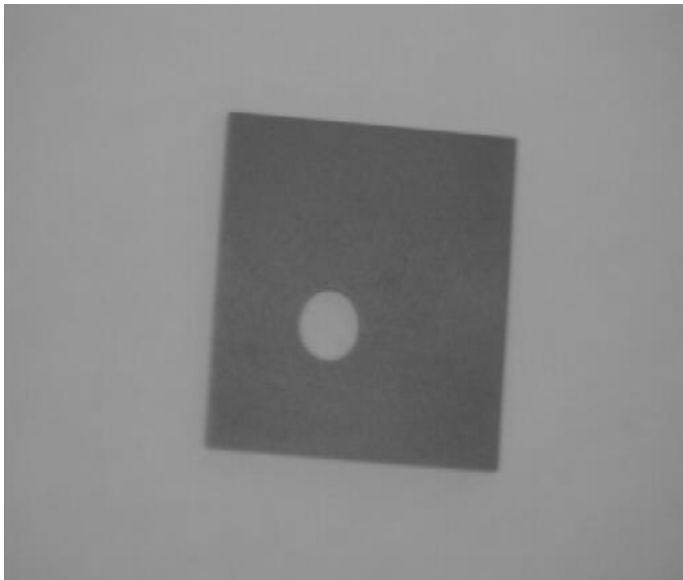
Global Thresholding

- Single threshold value for entire image
- Fixed ?
- Automatic
 - Intensity histogram



Global Thresholding

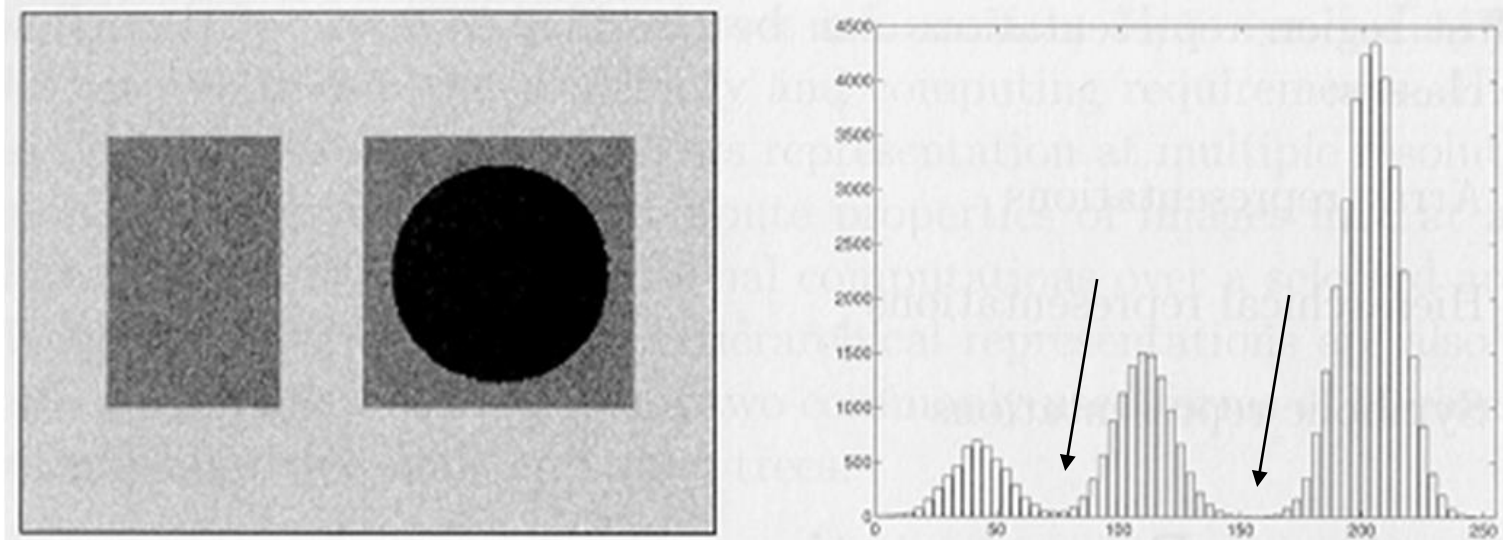
- Single threshold value for entire image
- Fixed ?
- Automatic
 - Intensity histogram



Global Thresholding

- Estimate an initial T
- Segment Image using T : Two groups of pixels $G1$ and $G2$
- Compute average gray values $m1$ and $m2$ of two groups
- Compute new threshold value $T = 1/2(m1 + m2)$
- Repeat steps 2 to 4 until: $\text{abs}(T_i - T_{i-1}) < \text{epsilon}$

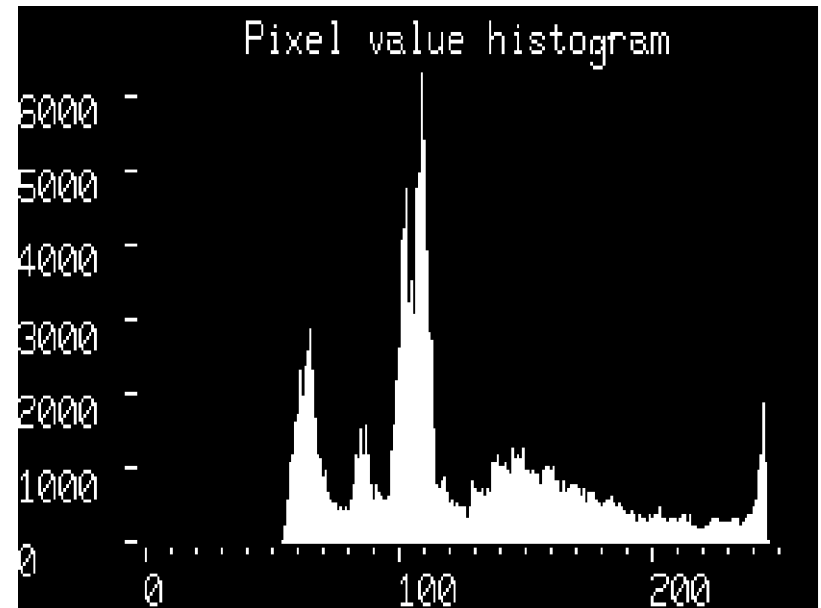
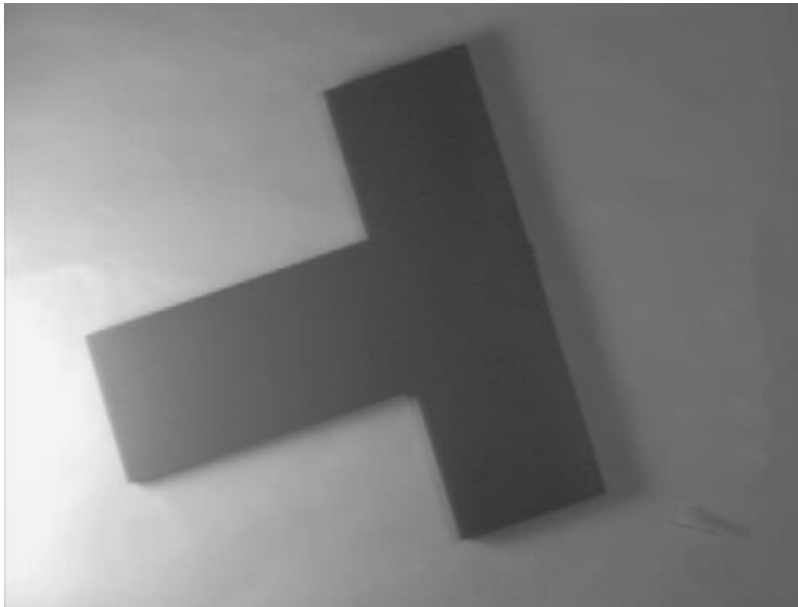
Global Thresholding



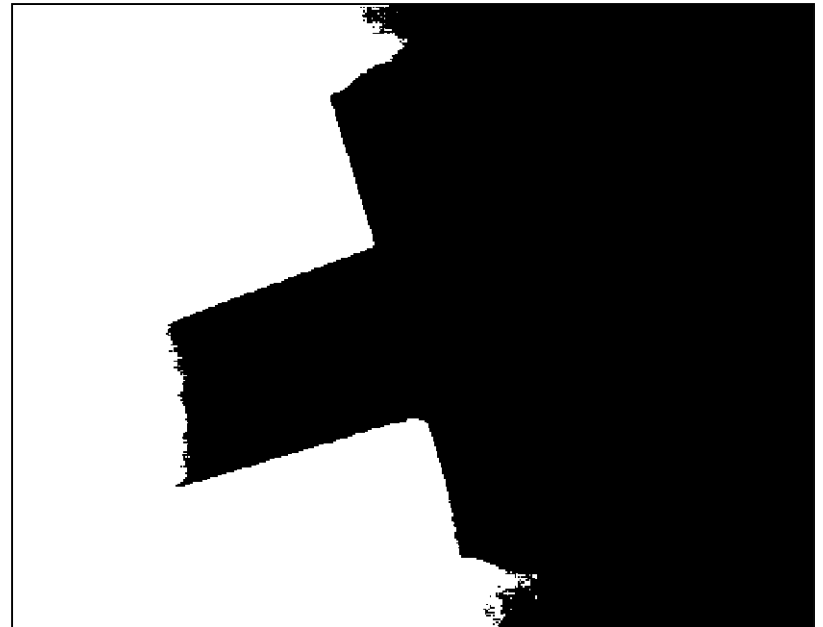
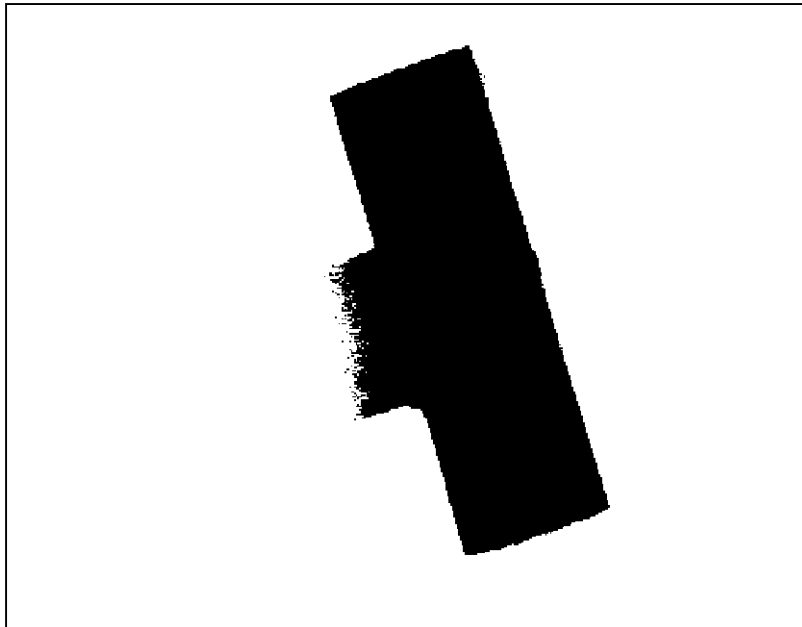
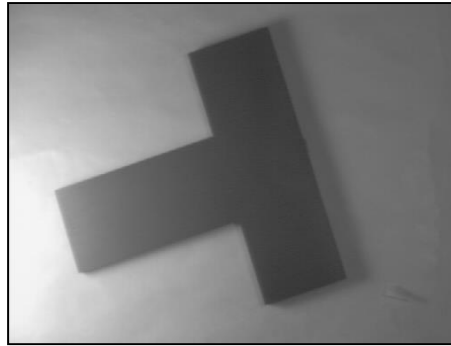
Multilevel thresholding

Thresholding

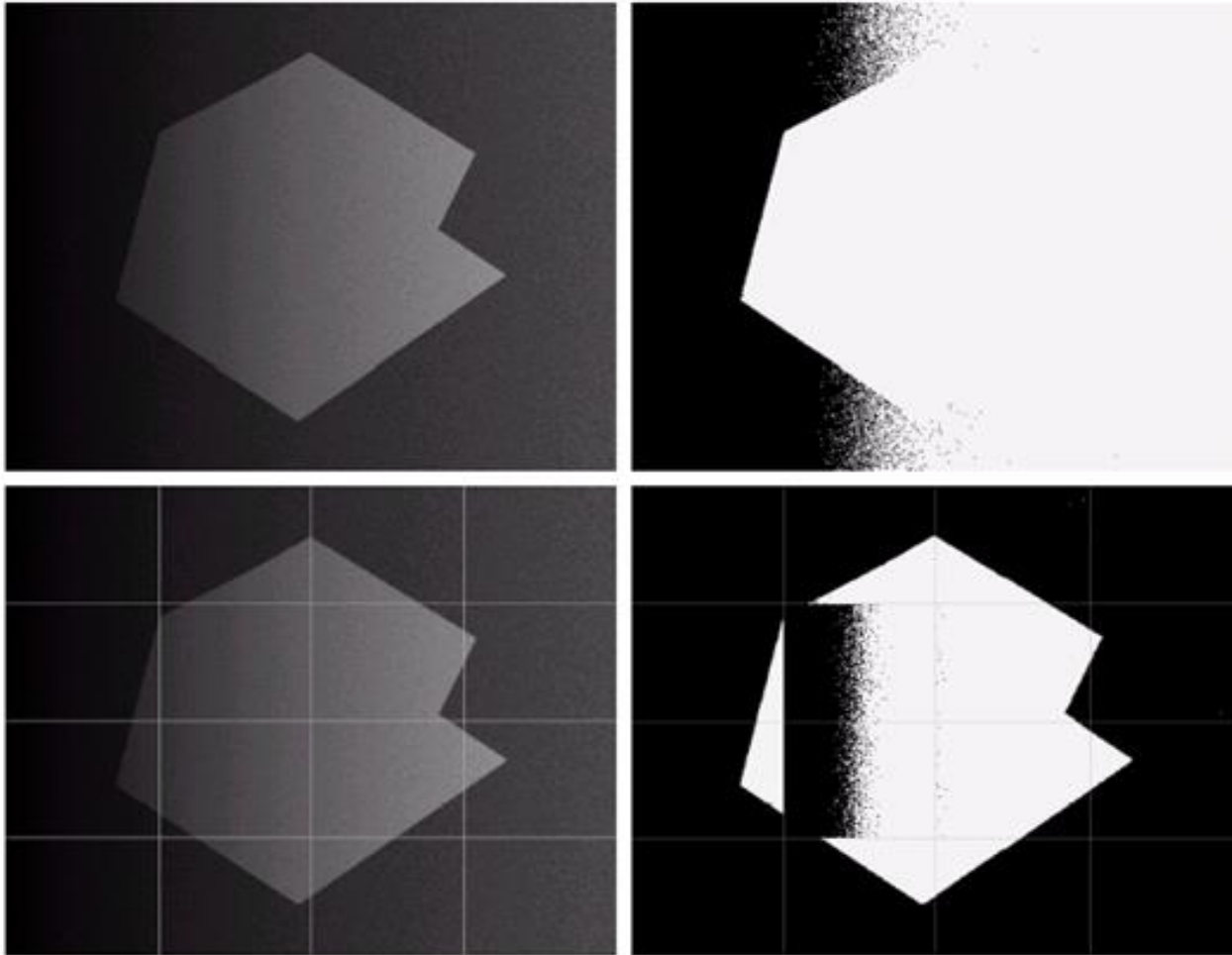
- Non-uniform illumination:



Global Thresholding



Adaptive Thresholding



Adaptive Thresholding

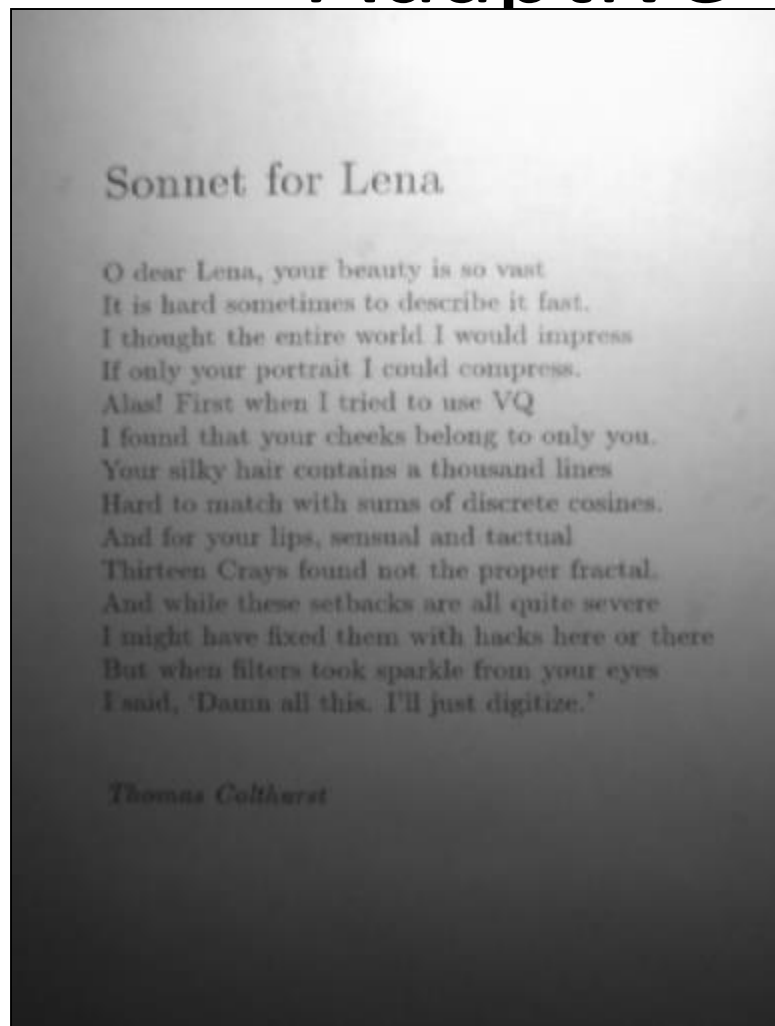
- Threshold: function of neighboring pixels

$$T = \textit{mean}$$

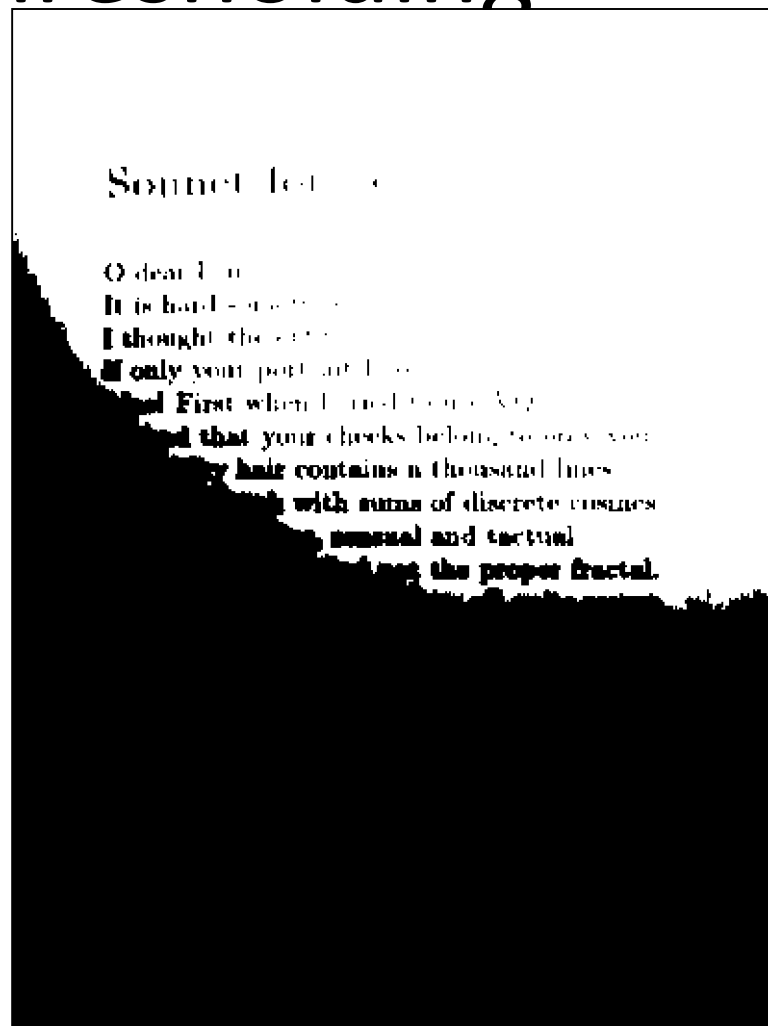
$$T = \textit{median}$$

$$T = \frac{\text{max} + \text{min}}{2}$$

Adaptive Thresholding

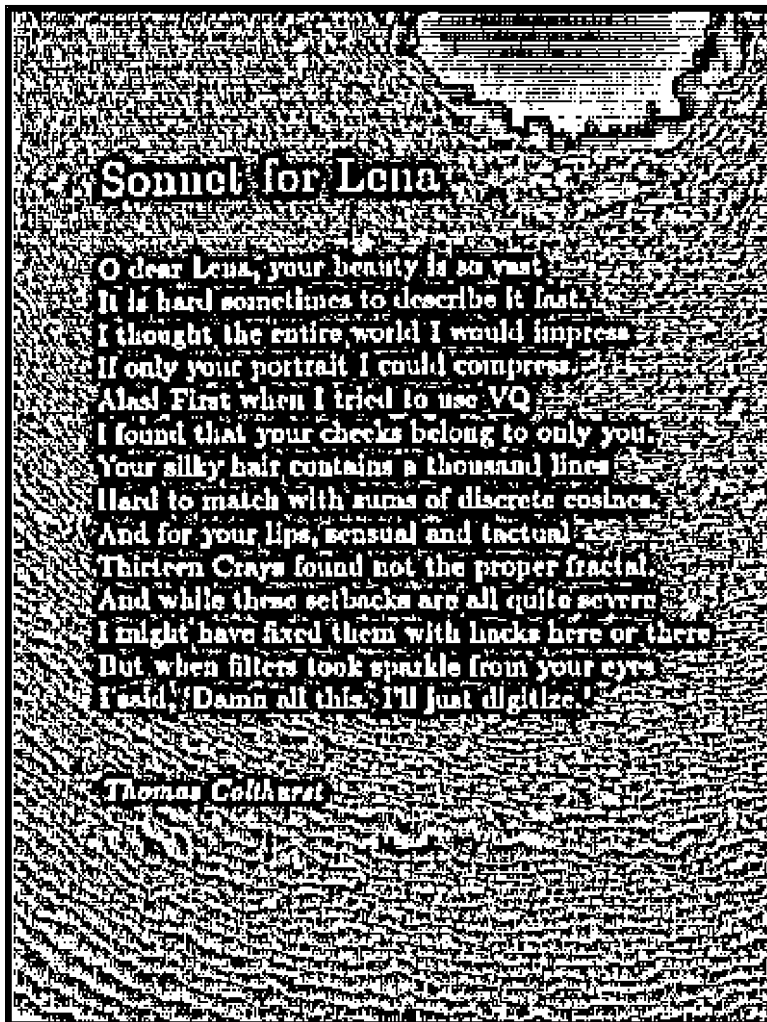


Original Image

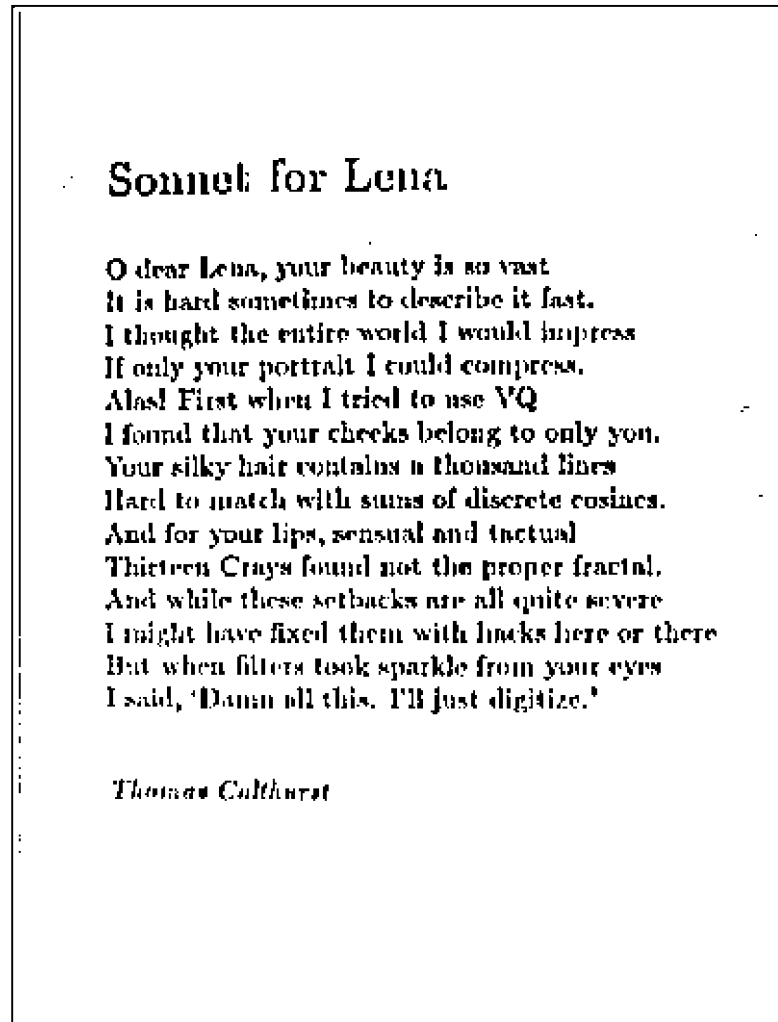


Global Thresholding

Adaptive Thresholding



T=mean, neighborhood=7x7



T=mean-Const., neighborhood=7x7

Adaptive Thresholding

- Niblack Algorithm

$$T = m + k \times s$$

m = mean

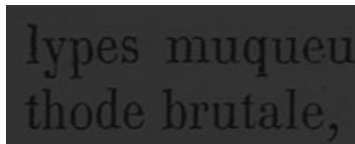
s = standard deviations

k = Niblack constant

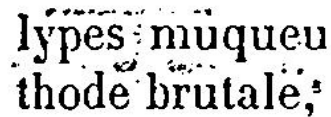
- Neighborhood size???

Document Binarization

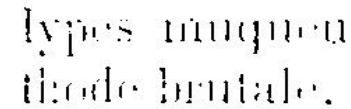
- Local Thresholding – Examples

The image shows the original document text, which is a snippet of a document with a dark background and light-colored text. The text is "lypes muqueu thode brutale,".

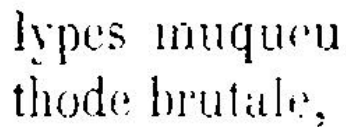
Original

The image shows the result of Niblack thresholding applied to the original text. The text is "lypes muqueu thode brutale,". The background is white, and the text is black, with some noise visible.

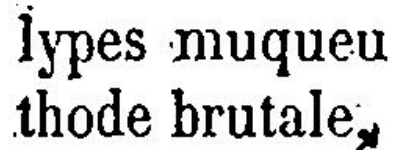
Niblack

The image shows the result of Sauvola thresholding applied to the original text. The text is "lypes muqueu thode brutale,". The background is white, and the text is black, with some noise visible.

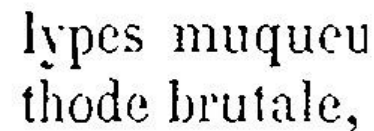
Sauvola

The image shows the result of Wolf thresholding applied to the original text. The text is "lypes muqueu thode brutale,". The background is white, and the text is black, with some noise visible.

Wolf

The image shows the result of Feng thresholding applied to the original text. The text is "lypes muqueu thode brutale,". The background is white, and the text is black, with some noise visible.

Feng

The image shows the result of NICK thresholding applied to the original text. The text is "lypes muqueu thode brutale,". The background is white, and the text is black, with some noise visible.

NICK

Region-Based Segmentation

- Divide the image into regions
 - $R_1, R_2, \dots, R_N$
- Following properties must hold:

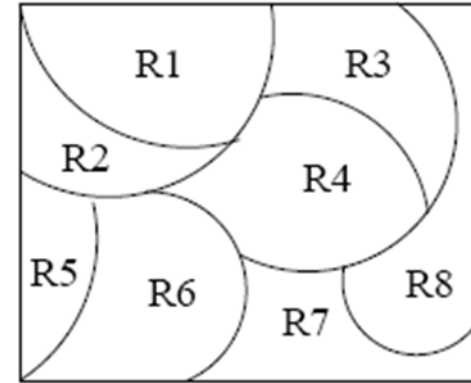
$$(1) R_1 \cup R_2 \cup \dots \cup R_n = R$$

(2) R_i is connected

$$(3) R_i \cap R_j = \text{empty}$$

$$(4) P(R_i) = \text{True}$$

$$(5) P(R_i \cup R_j) = \text{False} \quad (\text{For adjacent regions})$$



Region-Based Segmentation

■ Region Growing

- Region growing: groups pixels or subregions into larger regions.
- *Pixel aggregation*: starts with a set of “seed” points and from these grows regions by appending to each seed points those neighboring pixels that have similar properties (such as gray level).

1. Choose the seed pixel(s).
2. Check the neighboring pixels and add them to the region if they are similar to the seed
3. Repeat step 2 for each of the newly added pixels; stop if no more pixels can be added

Predicate: for example $\text{abs}(z_j - \text{seed}) < \text{Epsilon}$

Region-Based Segmentation

- Example

10	10	10	10	10	10	10
10	10	10	69	70	10	10
59	10	60	64	59	56	60
10	59	10	<u>60</u>	70	10	62
10	60	59	65	67	10	65
10	10	10	10	10	10	10
10	10	10	10	10	10	10

Region-Based Segmentation

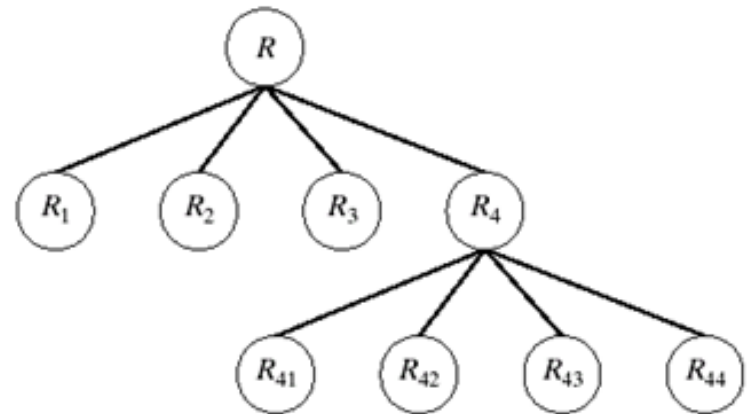
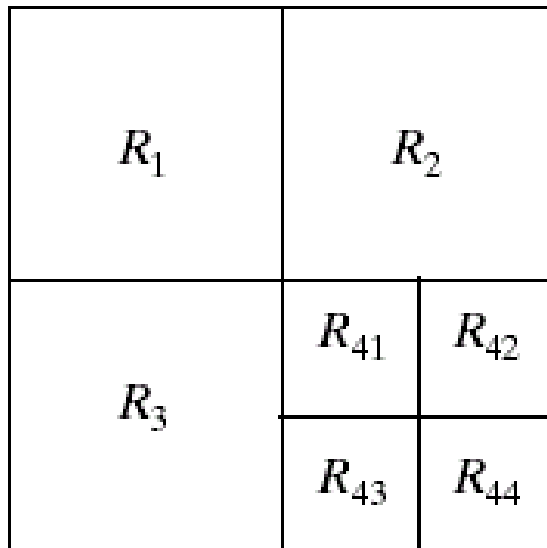
■ Region Splitting

- Region Growing: Starts from a set of seed points.
- Region Splitting: Starts with the whole image as a single region and subdivide the regions that do not satisfy a condition.
- Image = One Region R
- Select a predicate P (gray values etc.)
- Successively divide each region into smaller and smaller quadrant regions so that:

$$P(R_i) = \text{true}$$

Region-Based Segmentation

- Region Splitting



Problem? Adjacent regions could be same

Solution? Allow Merge

Region-Based Segmentation

- Region Merging

- Region merging is the opposite of region splitting.
- Merge adjacent regions R_i and R_j for which:

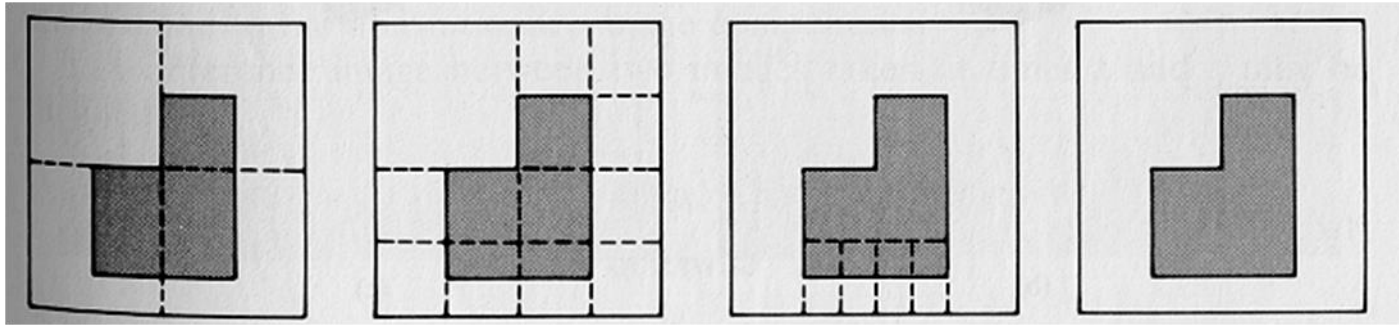
$$P(R_i \cup R_j) = \text{True}$$

- Region Splitting/Merging

- Stop when no further split or merge is possible

Region-Based Segmentation

■ Example

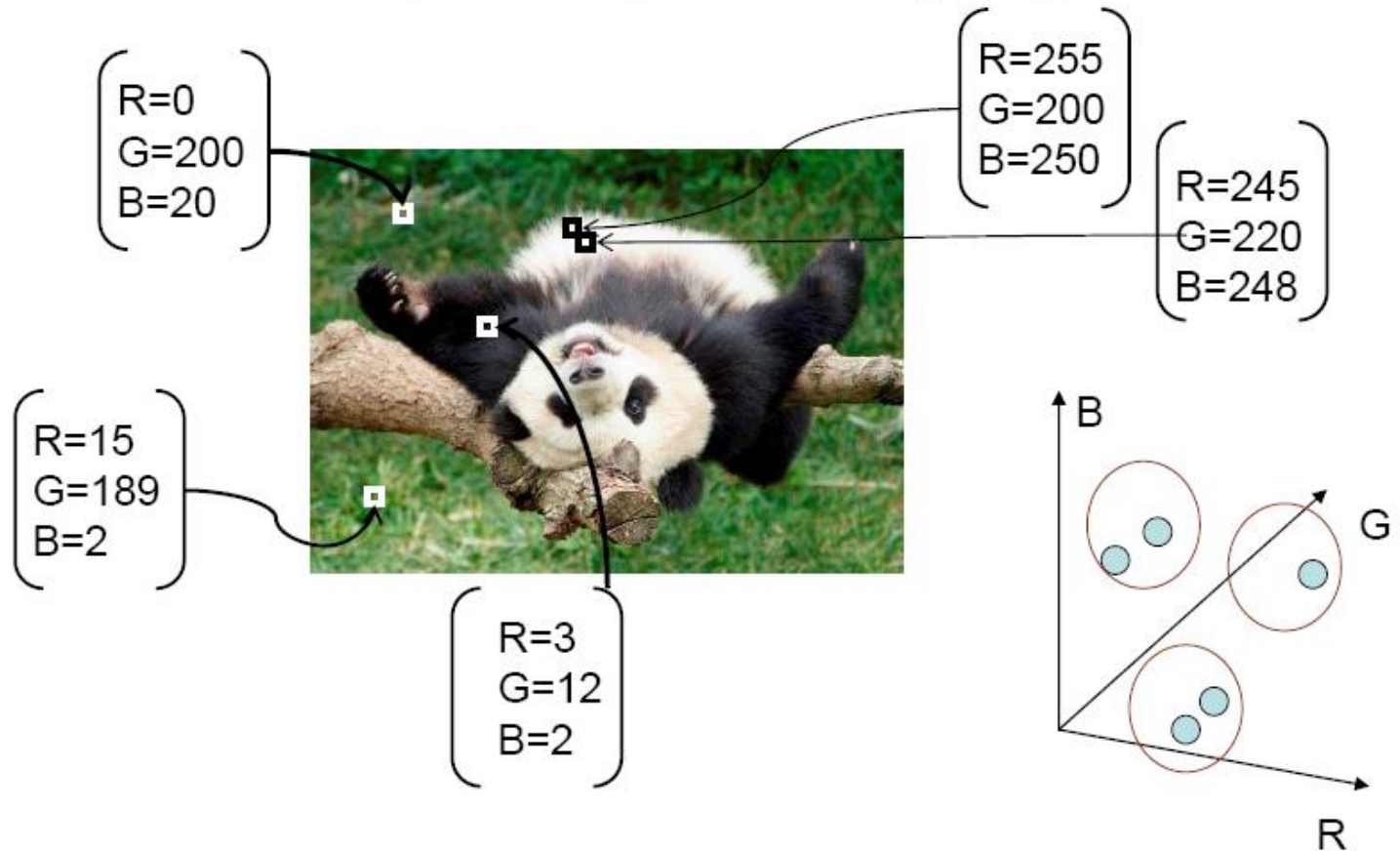


1. Split into four disjointed quadrants any region R_i where $P(R_i)=\text{False}$
2. Merge any adjacent regions R_j and R_k for which $P(R_j \cup R_k)=\text{True}$
3. Stop when no further merging or splitting is possible

Cluster Analysis

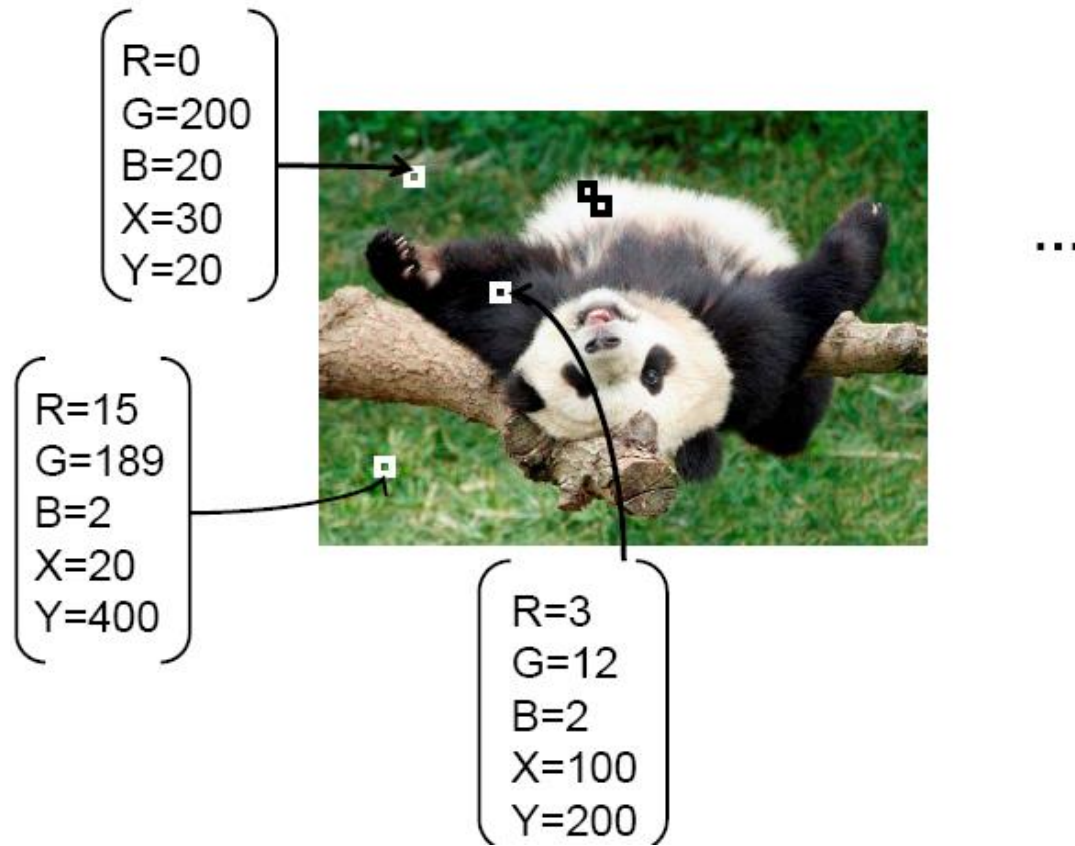
Segmentation as clustering

- Cluster similar pixels (features) together



Segmentation as clustering

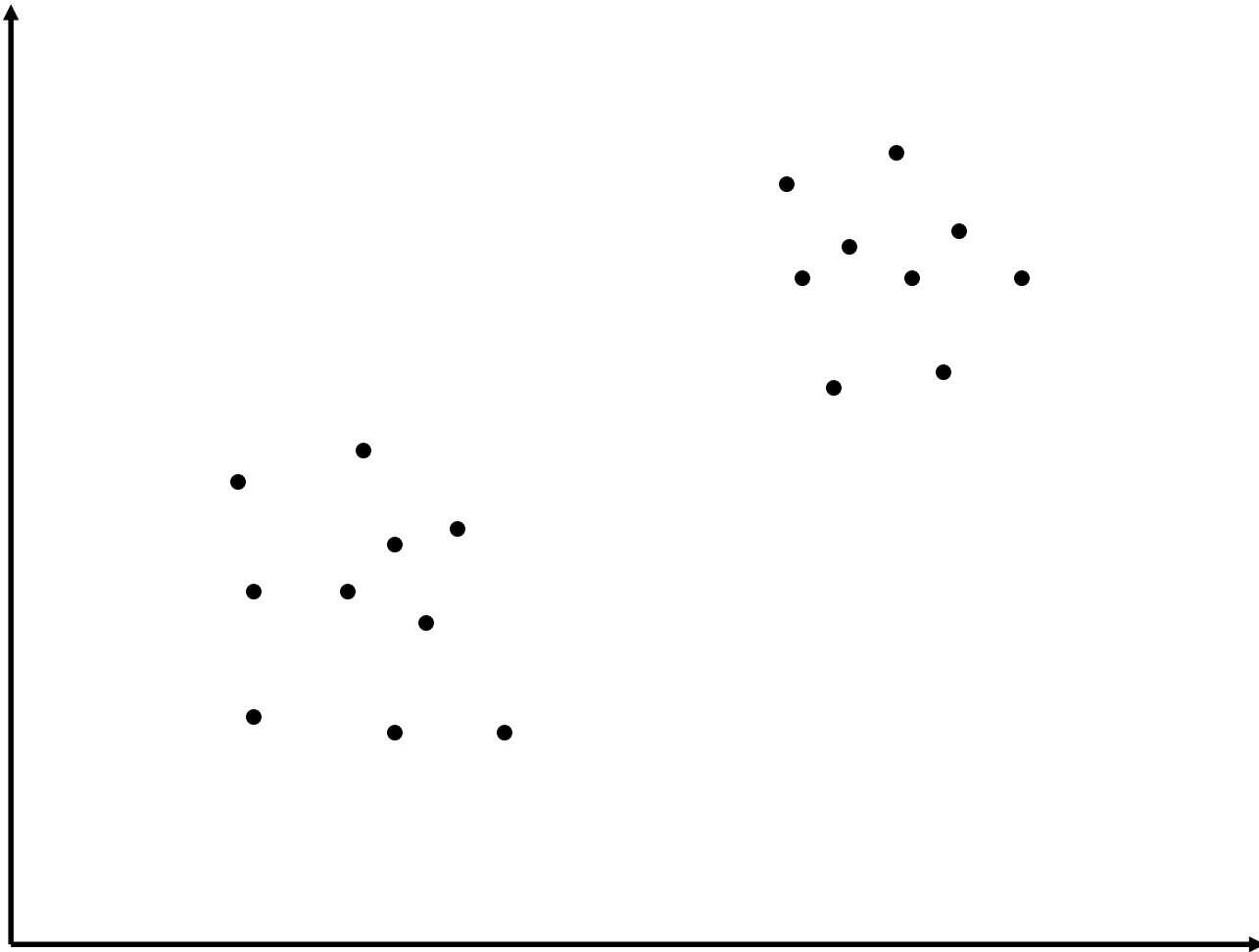
- Cluster similar pixels (features) together



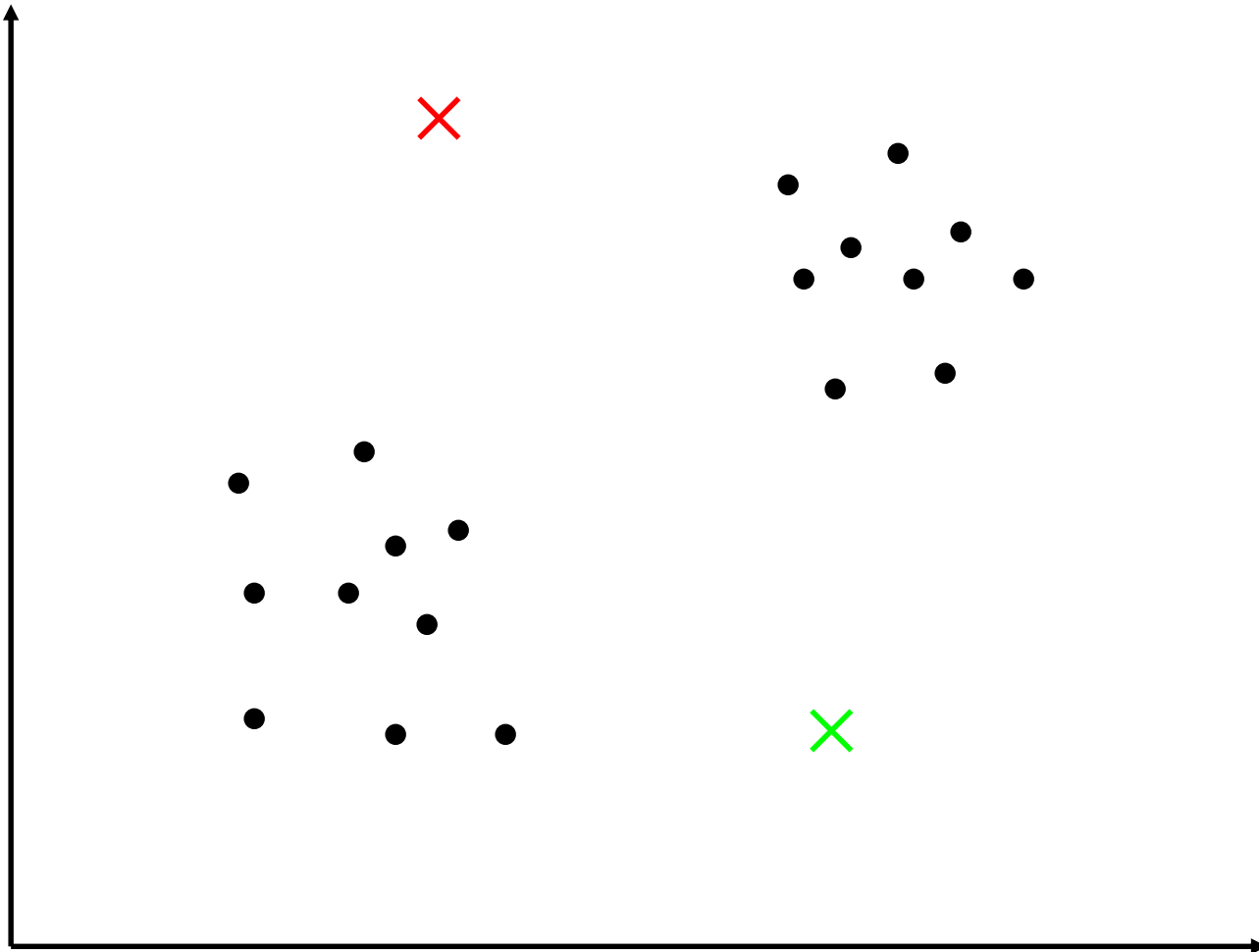
K-Means Clustering

1. Chose the number (K) of clusters and randomly select the centroids of each cluster.
2. For each data point:
 - ❑ Calculate the distance from the data point to each cluster.
 - ❑ Assign the data point to the closest cluster.
3. Recompute the centroid of each cluster.
4. Repeat steps 2 and 3 until there is no further change in the assignment of data points (or in the centroids).

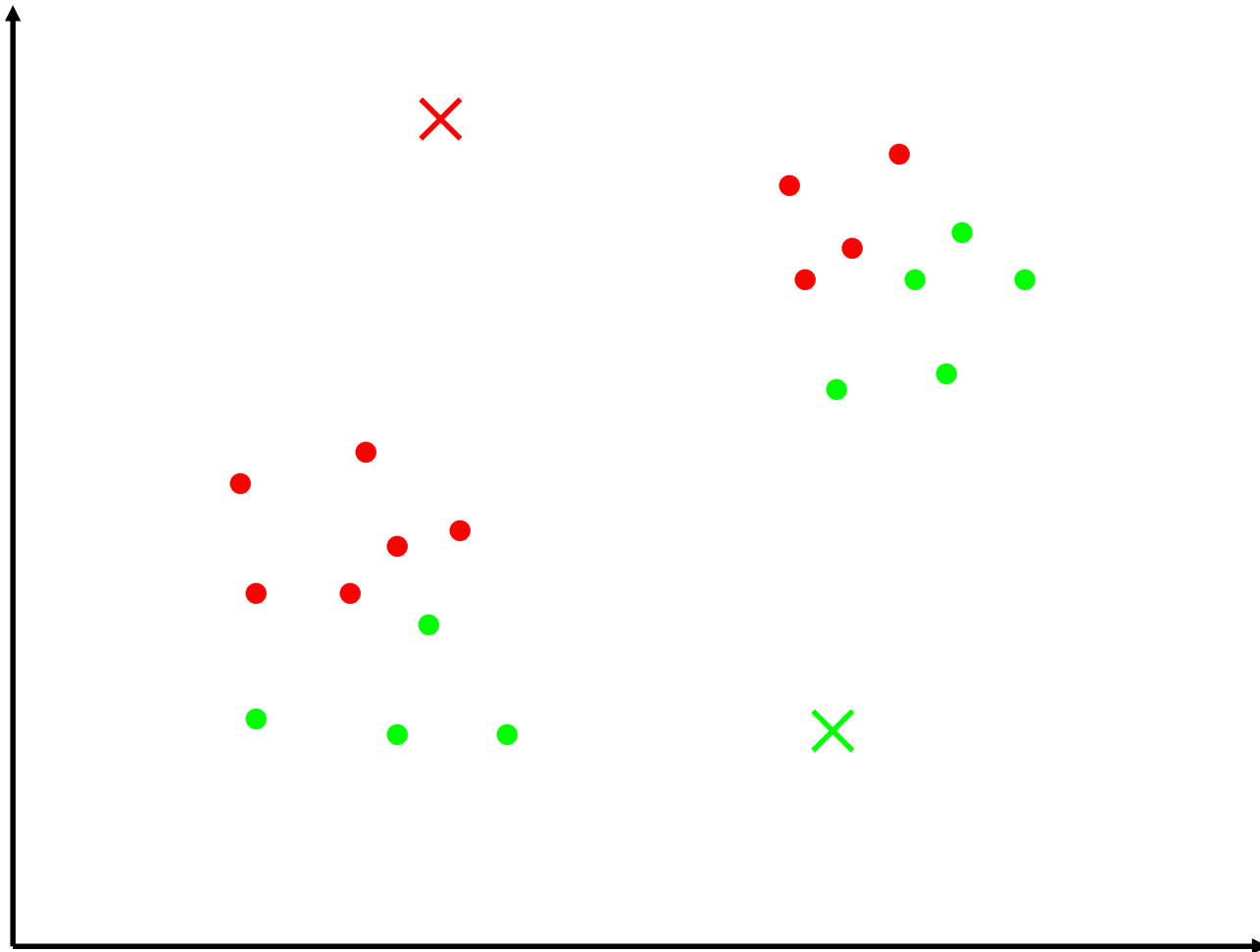
K-Means Clustering



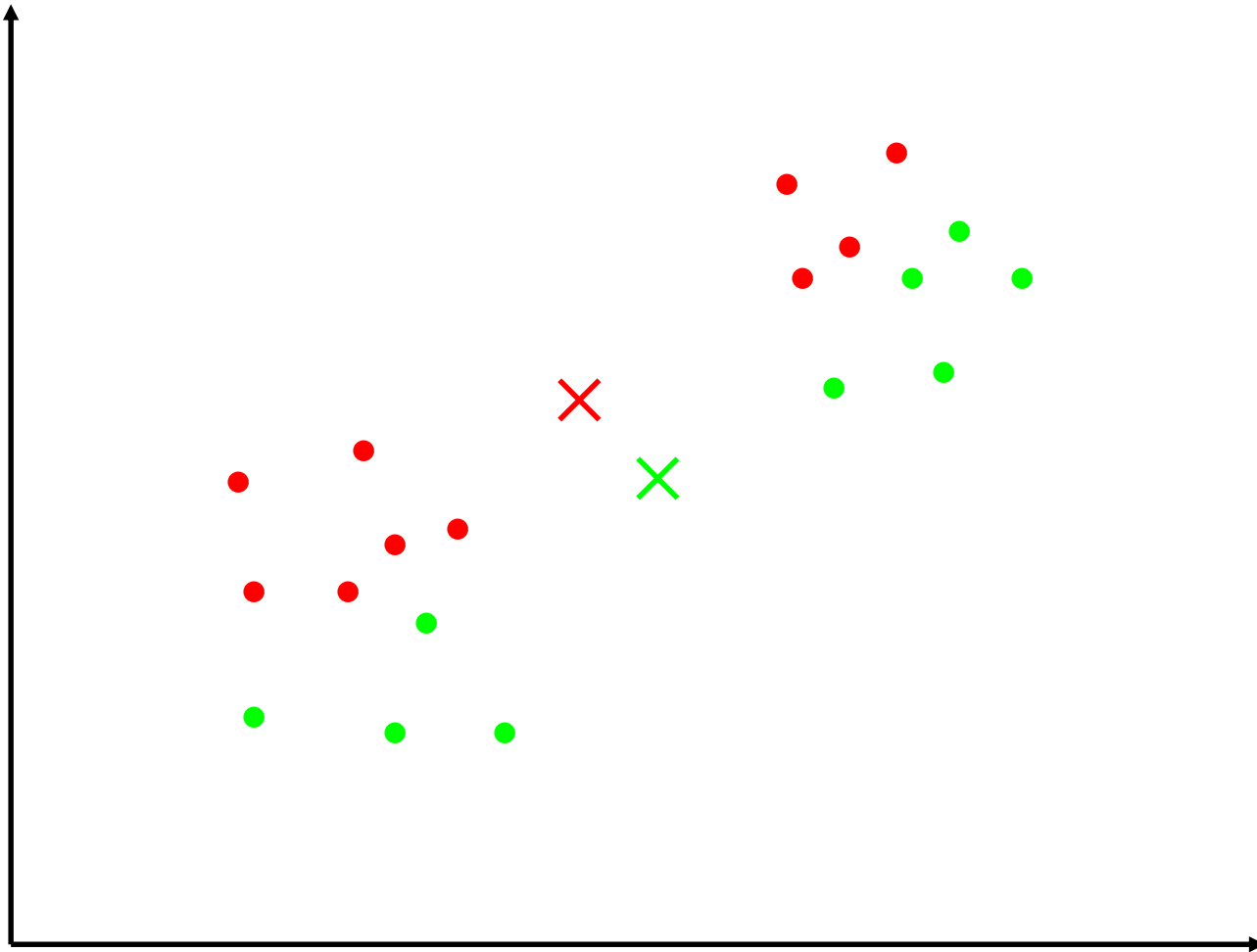
K-Means Clustering



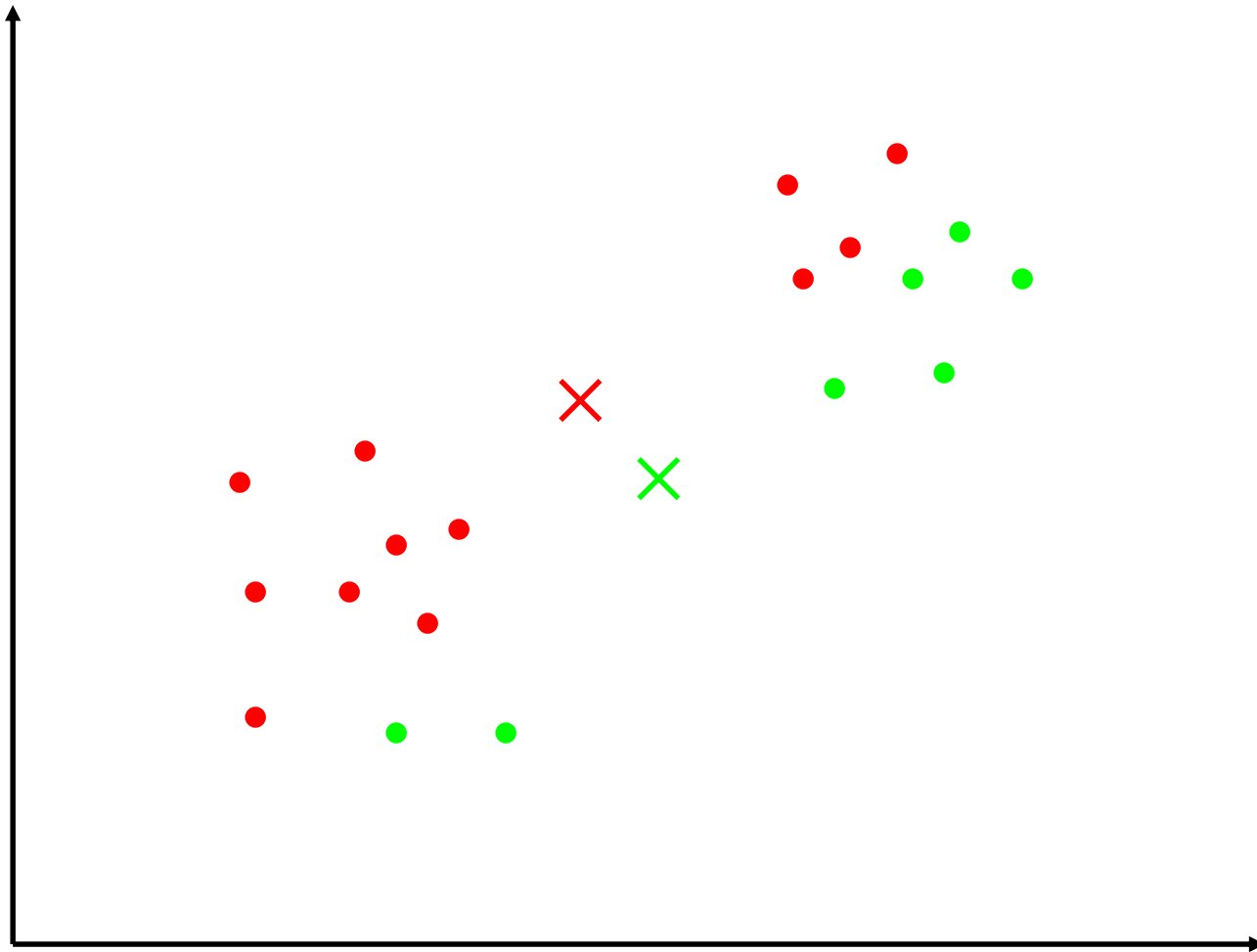
K-Means Clustering



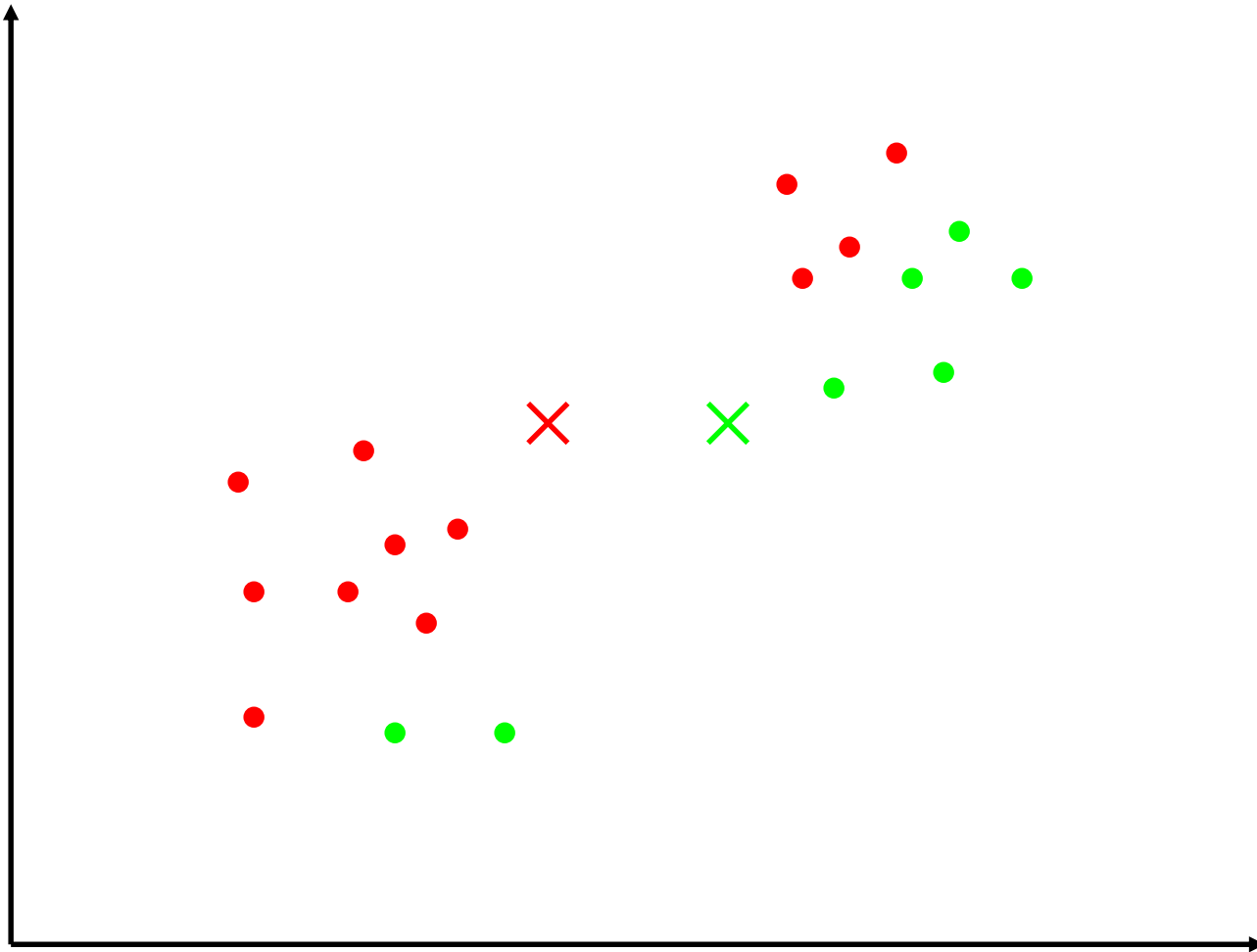
K-Means Clustering



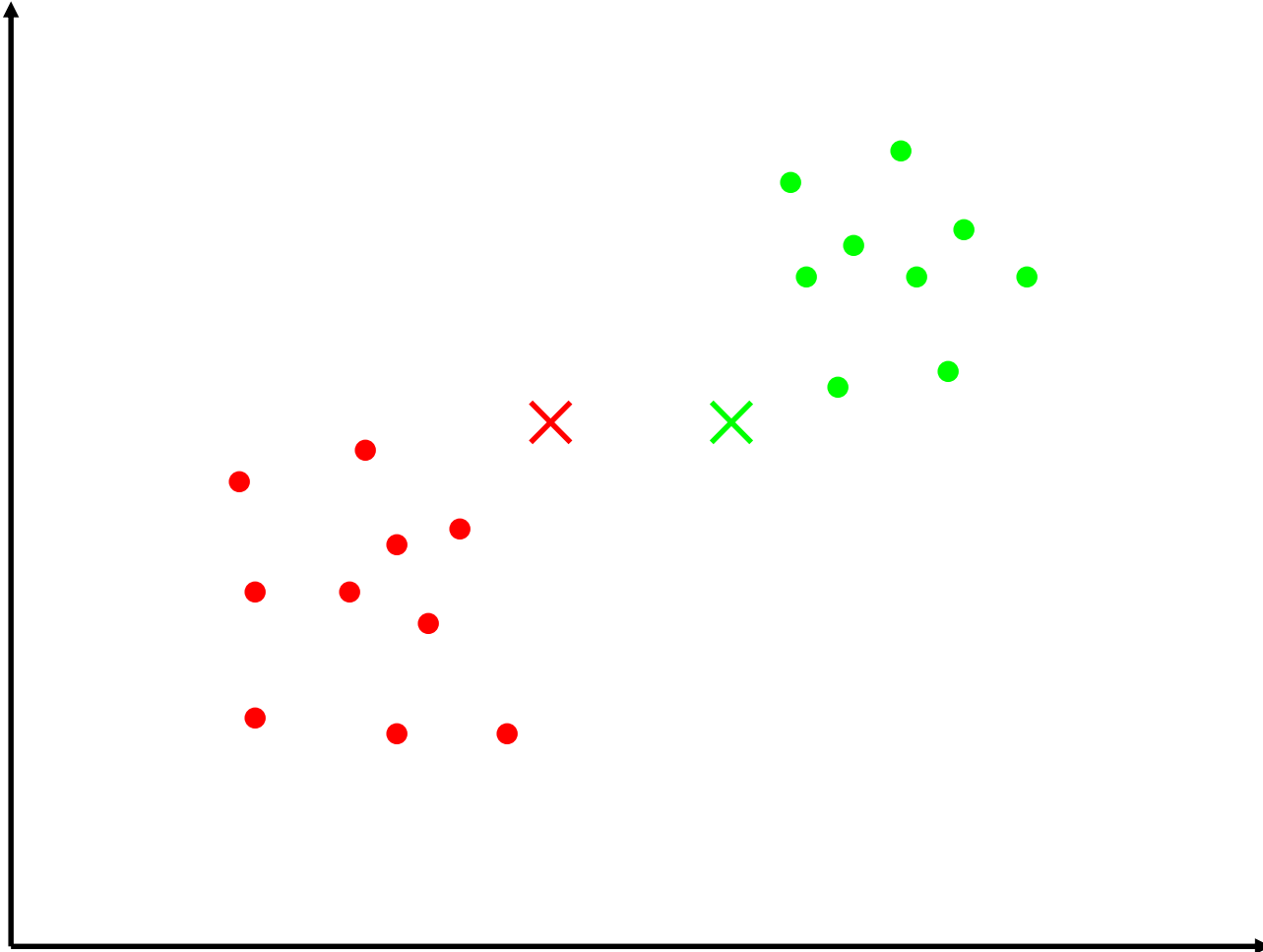
K-Means Clustering



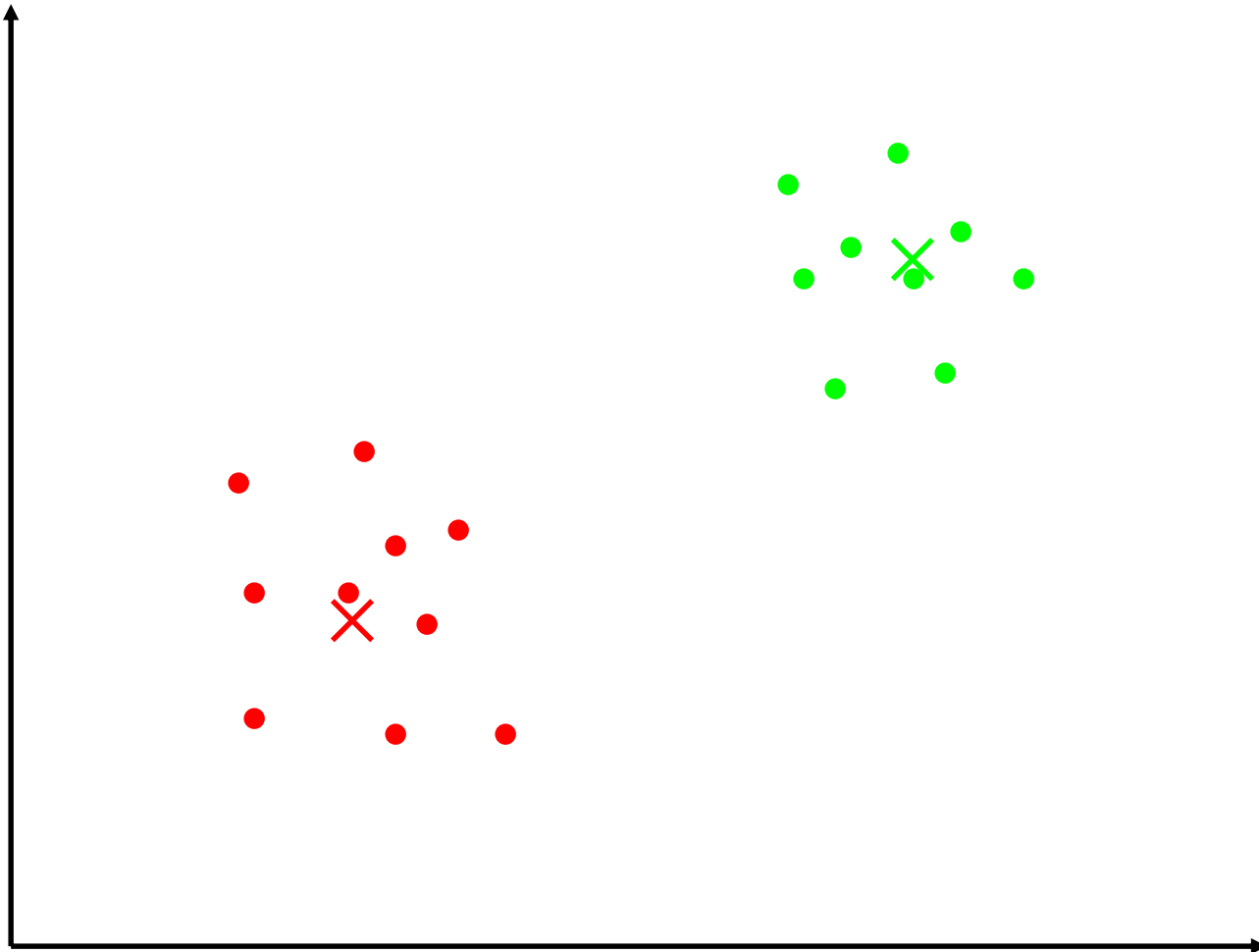
K-Means Clustering



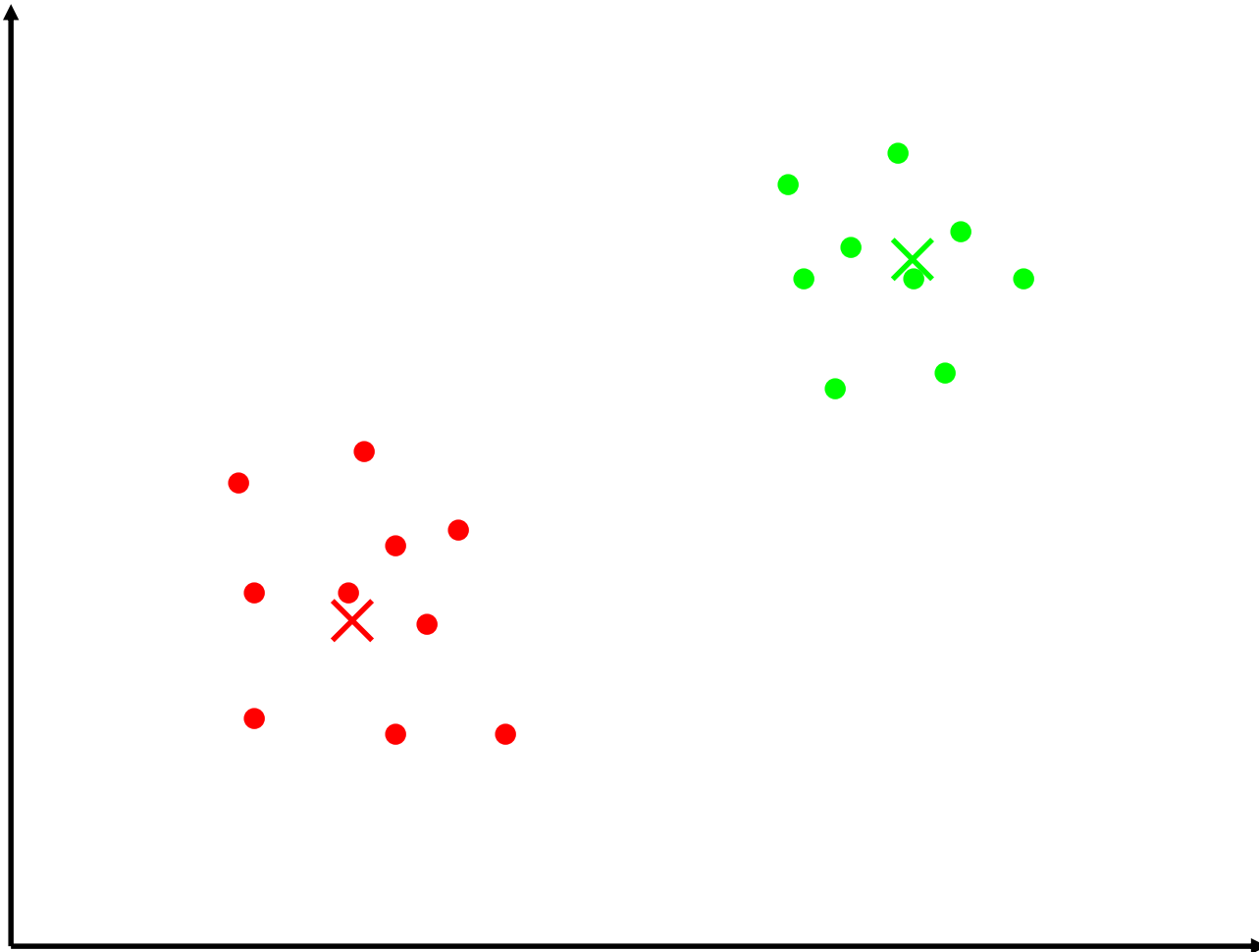
K-Means Clustering



K-Means Clustering

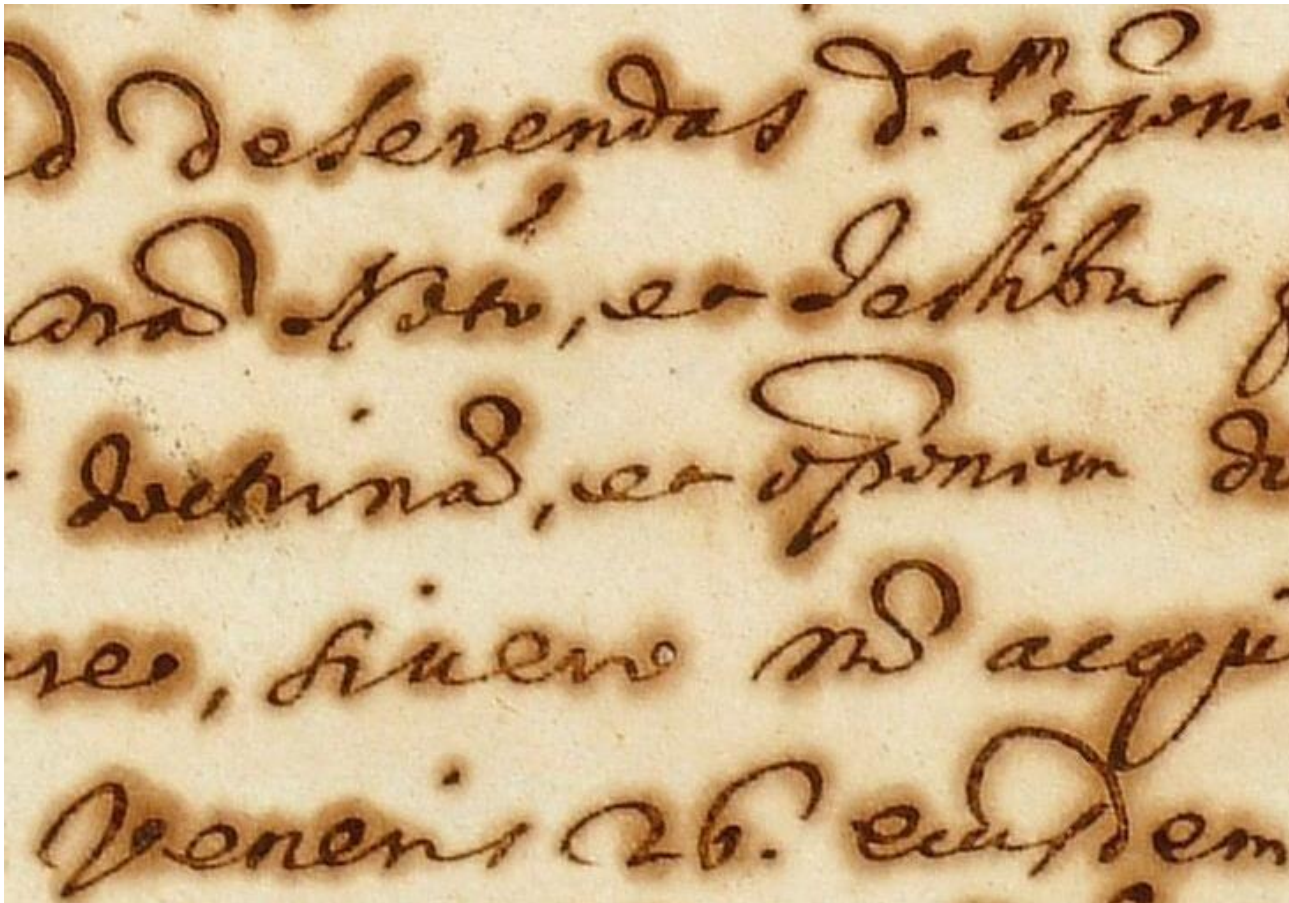


K-Means Clustering



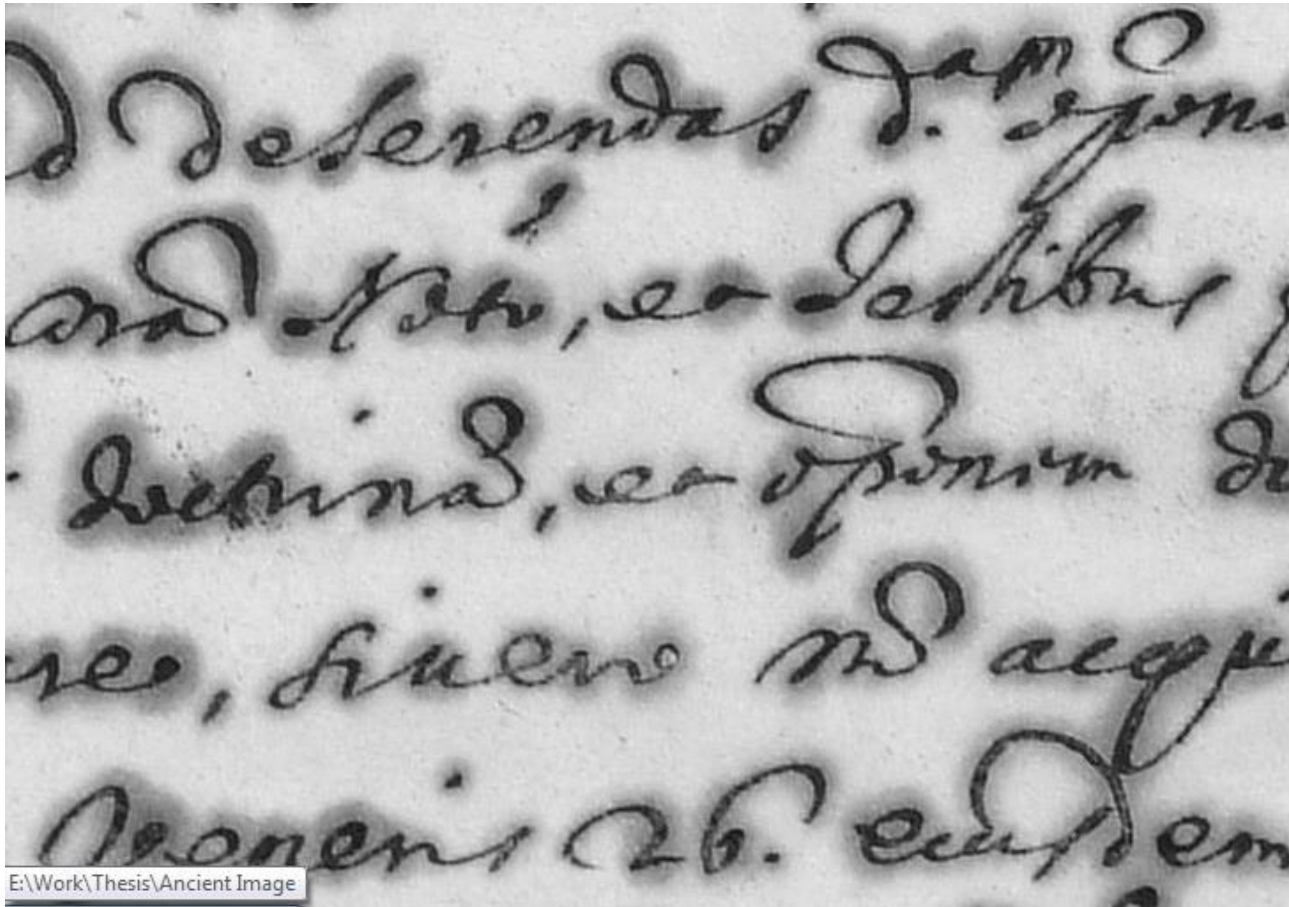
Clustering

- Example



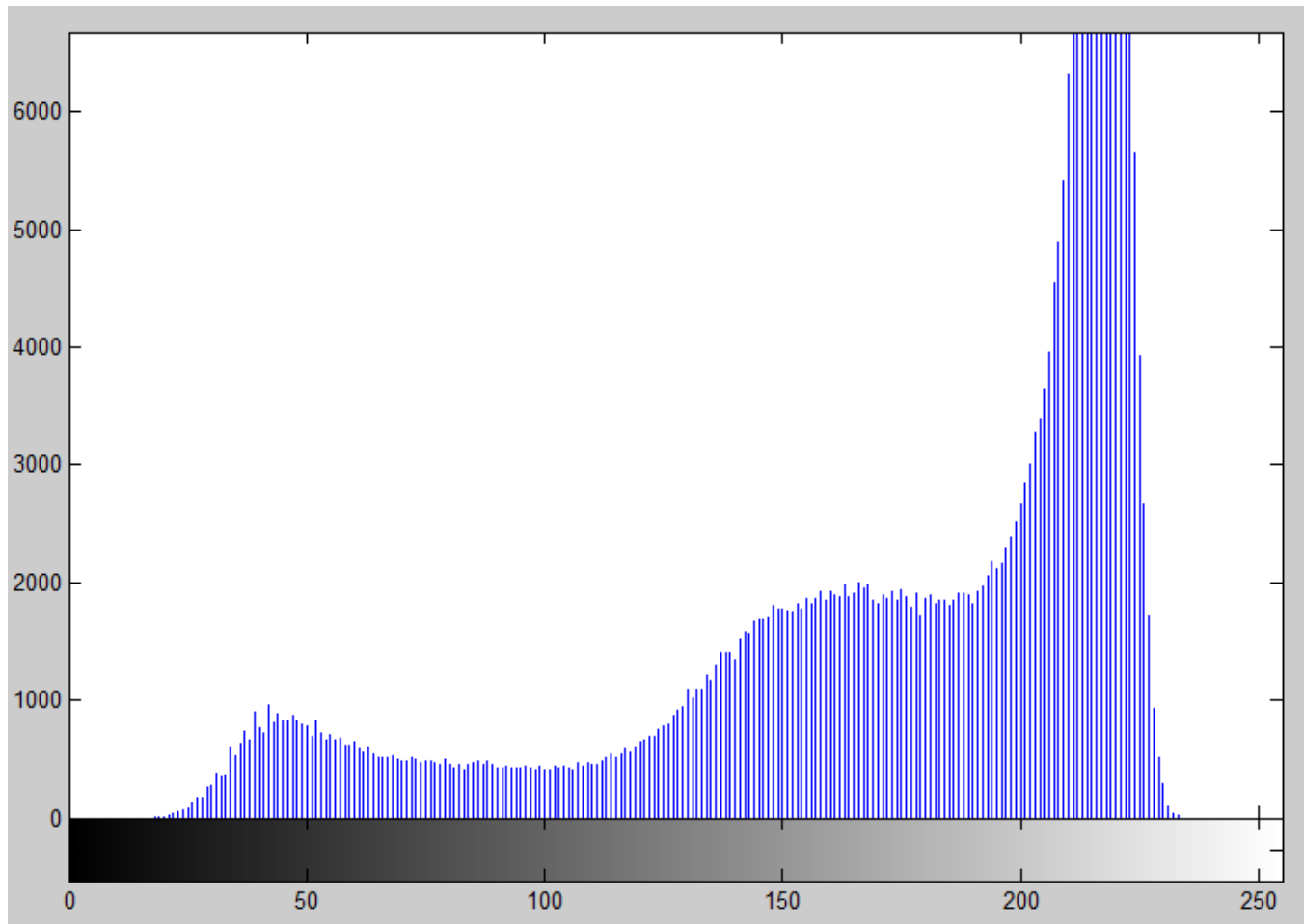
Clustering

- Example



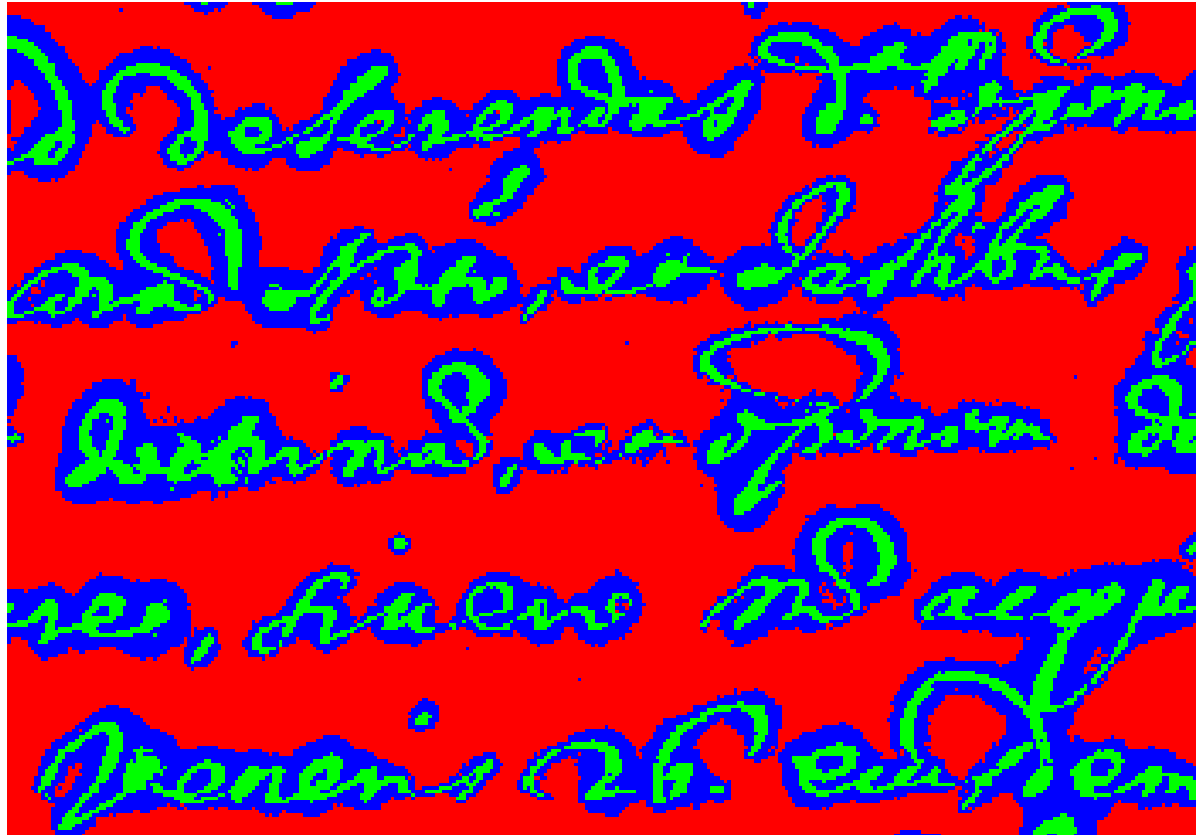
Clustering

- Example



Clustering

- Example



Clustering

- Example



D. Comaniciu and P. Meer, *Robust Analysis of Feature Spaces: Color Image Segmentation*, 1997.

K-Means Clustering

- Example



Original



K=5



K=11

Acknowledgements

- ◆ Digital Image Processing”, Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- ◆ Computer Vision for Computer Graphics, Mark Borg
- ◆ Computer Vision A modern Approach by Frosyth
- ◆ CSCI 1430: Introduction to Computer Vision by [James Tompkin](#)